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Review the assignment due date

Final Project

Team Number: 5

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GitHub Repository URL: https://github.com/upenn-embedded/final-project-s26-t5

GitHub Pages Website URL: [for final submission]*

Final Project Proposal

1. Abstract

This project presents a prototype for volumetric electromagnetic control targeting optimal remediation (VECTOR) to manipulate magnetic nanoparticles (MNPs) using a 3-by-5 array of individually addressable electromagnets driven by an ATmega328PB microcontroller. The system is designed to demonstrate programmable spatial control of magnetic material through two operating modes: a time display mode, in which the current time is rendered as a magnetic pattern, and an animation mode in which input patterns are generated across the array. A Hall-effect sensing subsystem provides feedback to verify field operation, while an LCD and button interface enables real-time user interaction.

2. Motivation

Water contamination remains a major engineering challenge, and many conventional treatment methods can be energy-intensive, chemically harmful, or difficult to deploy in preexisting systems. Ferric-oxide MNPs are interesting as they offer a platform to functionalize polymers, which attach to contaminants and can be manipulated or recovered using magnetic fields. If such particles can be moved in a controlled manner, future systems could use them to target the collection, transport, and separation of harmful substances from water.

The recent deregulation by President Trump and Administrator Zeldin on February 12, 2026, has been considered the single largest deregulatory action in U.S. history. The Mercury and Air Toxics Standards (MATS) is a program that successfully reduced mercury

emissions by 90% prior to its repeal, and its repeal has immediate, quantifiable effects on waterway pollution. Other toxins, such as arsenic and lead, whose limits have been repealed, will also plateau or reverse pollution reductions achieved since 2012. Without federal limits on harmful practices, there is an expected 23% increase in mercury emissions, with 50% stemming from the dirtiest coal plants by 2035. Toxic metals in wastewater are projected to add another 325,000 tons of pollution to rivers and streams each year, further compromising already dwindling water sources.

Functionalized MNPs are considered an implementable, highly targeted solution to this growing issue. Yet it remains limited by a fundamental technical challenge: reliably controlling magnetic materials with an electronically programmable field source. Current practices of functionalized MNPs in water discard the magnetic material, making it inefficient and costly. To create a full remediation system, a platform demonstrating repeatable magnetic actuation is necessary to verify that particles can be retained and controlled for future use in waterways. Our prototype addresses that need by building a small-scale array of electromagnets that can generate visible, programmable particle motion.

The motivation of this project is not to fully solve water remediation, but to create an experimental stepping stone toward that goal. By showing that magnetic material can be directed into specific spatial patterns and moved according to programmed sequences, this prototype helps establish the feasibility of using electronically controlled electromagnets for future nanoparticle handling in research. In later iterations, this idea could be extended to benchtop demonstrations in fluidic-controlled environments, stronger field geometries, closed-loop sensing, and ultimately to systems that move MNPs in flowing water for contaminant capture and removal.

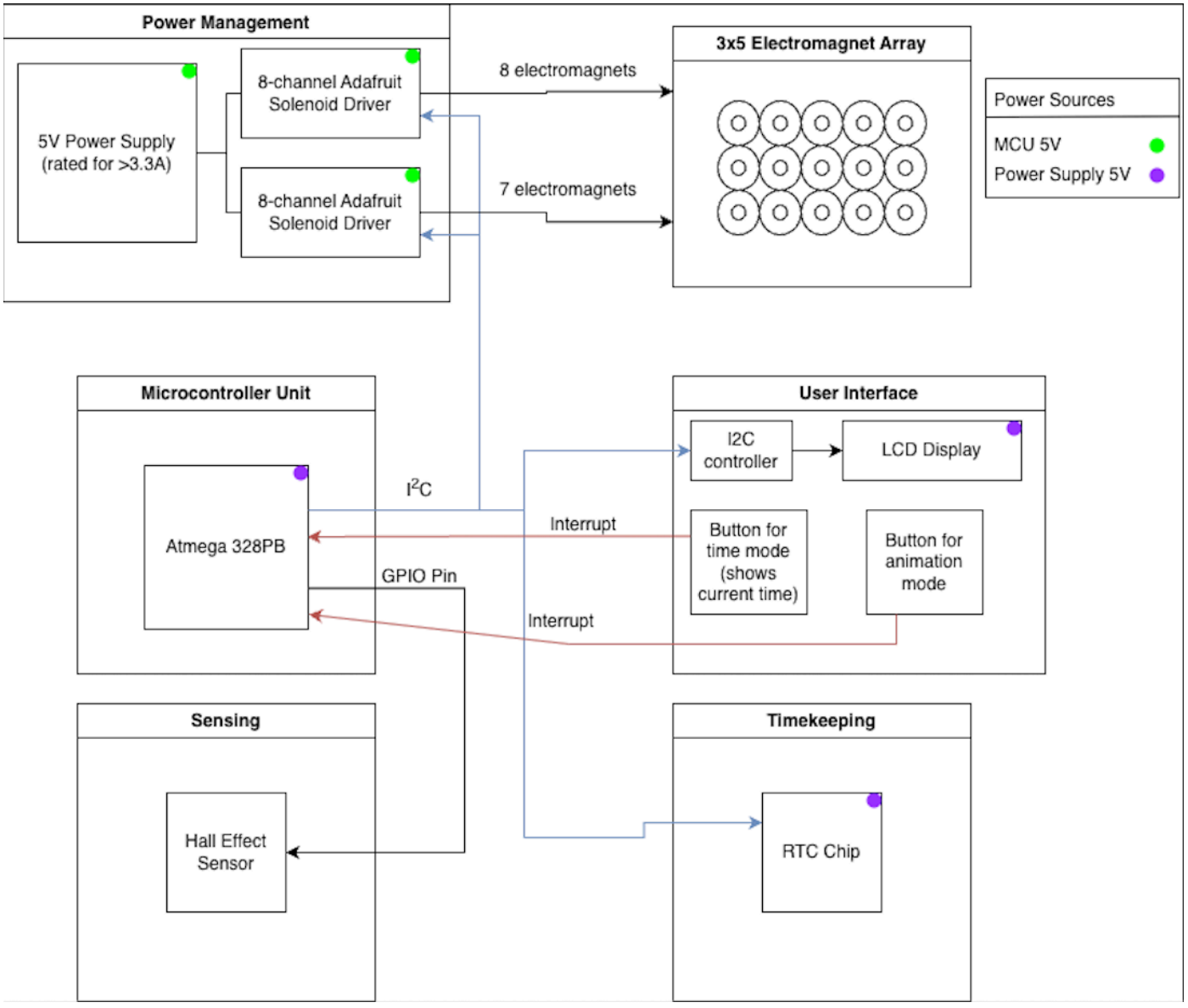
3. System Block Diagram

The most critical component in our project is the array of electromagnets. These will be arranged in a 3x5 grid to demonstrate directional control. The electromagnets are powered at 5V at a max current draw of 0.22A. Thus, we anticipate a max current draw of 3.3A for 15 electromagnets. The power for these electromagnets will be supplied via a power supply, most likely from the lab bench. First, power will be routed from the power supply to an 8-channel power management IC from Adafruit that is specialized for solenoids, featuring MOSFETs with flyback diodes. We plan to use two of these and to control them with I2C from an Atmega 328PB.

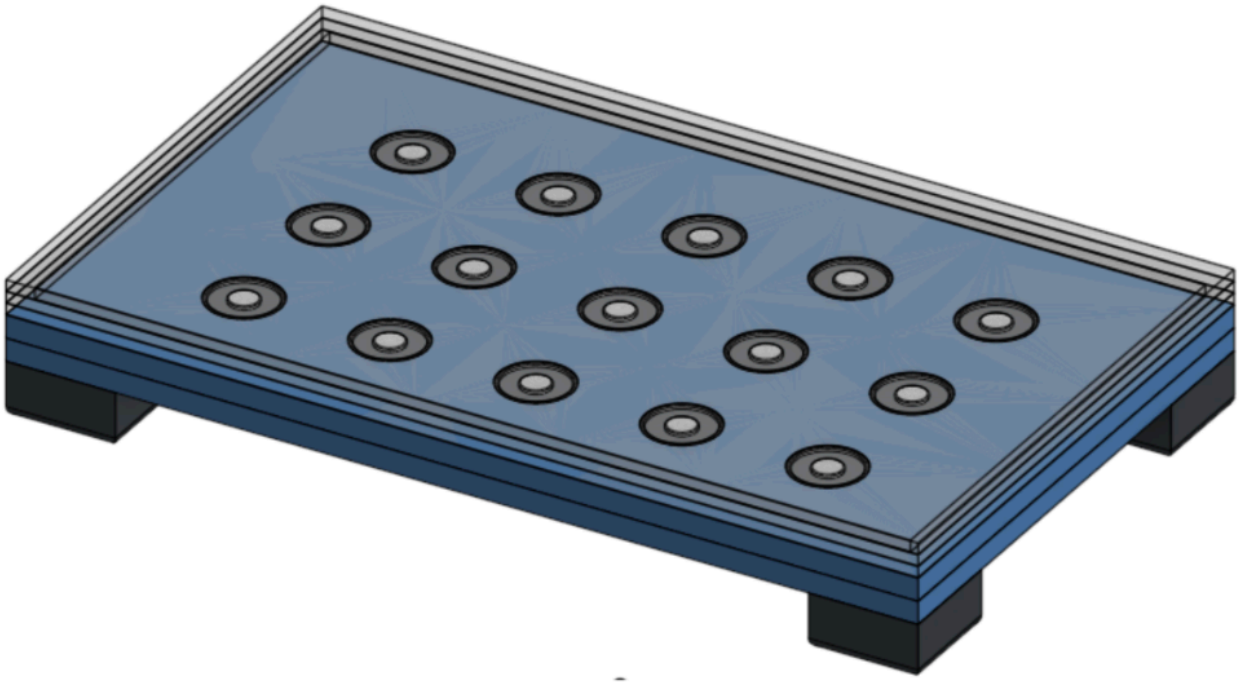
For an input device, we will have a Hall effect sensor probe to ensure that the electromagnetic array is operating properly. It will communicate via a GPIO pin with the MCU.

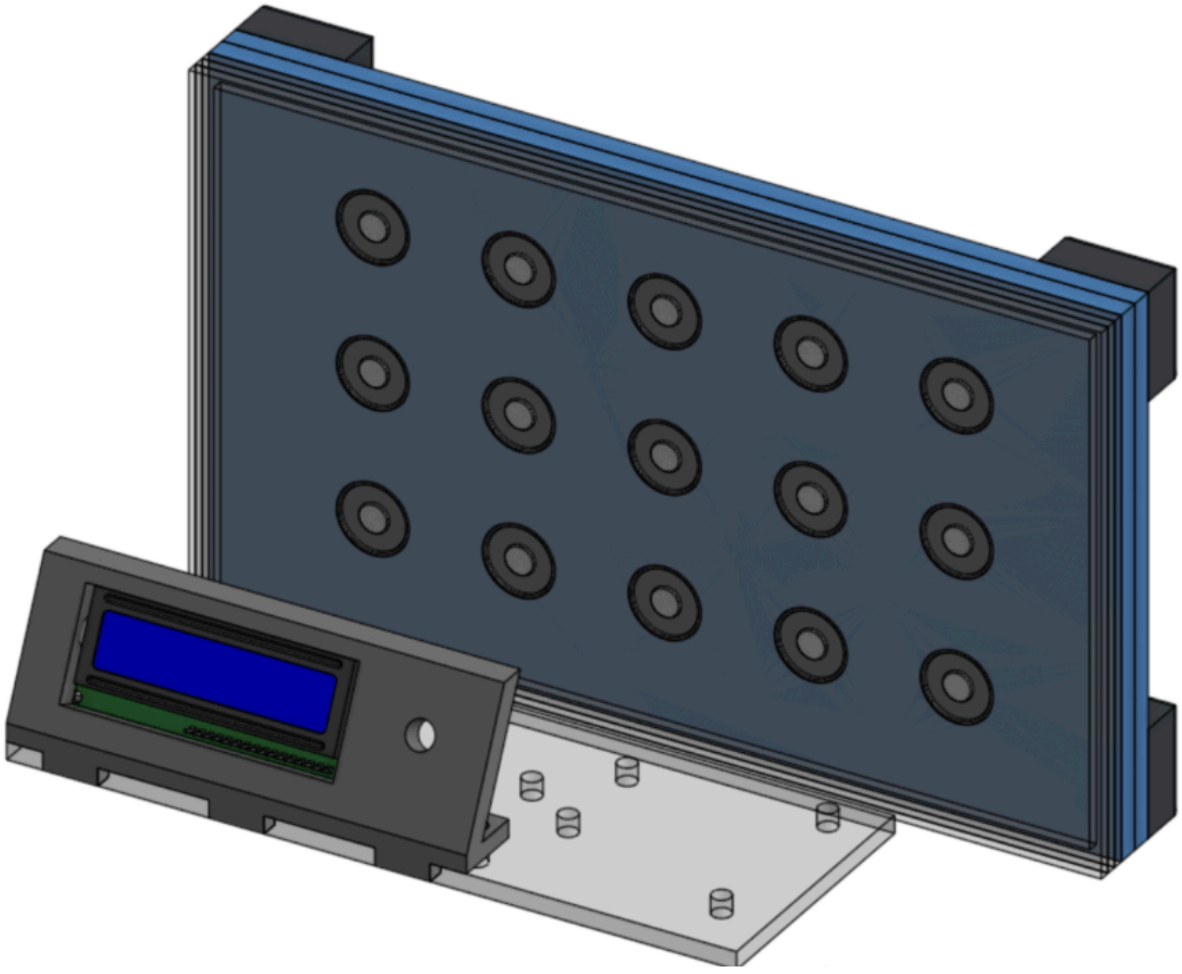
Next, we also have a user interface. Here, the user will be able to press two buttons to demonstrate two separate modes on the electromagnetic array. The first mode will be a clock mode, in which the current time will be spelled out digit-by-digit on the array. The second mode will be an animation mode, in which the array will animate a path of moving magnetic nanoparticles. Telemetry such as the mode and whether a magnetic field was detected by the Hall effect sensor will be displayed on an LCD. We will use an I2C backpack for this LCD, which simplifies communication.

Lastly, we will use a real-time clock (RTC) chip that can tell the time even when the system is not connected to the internet. This chip will use I2C to talk to the MCU.



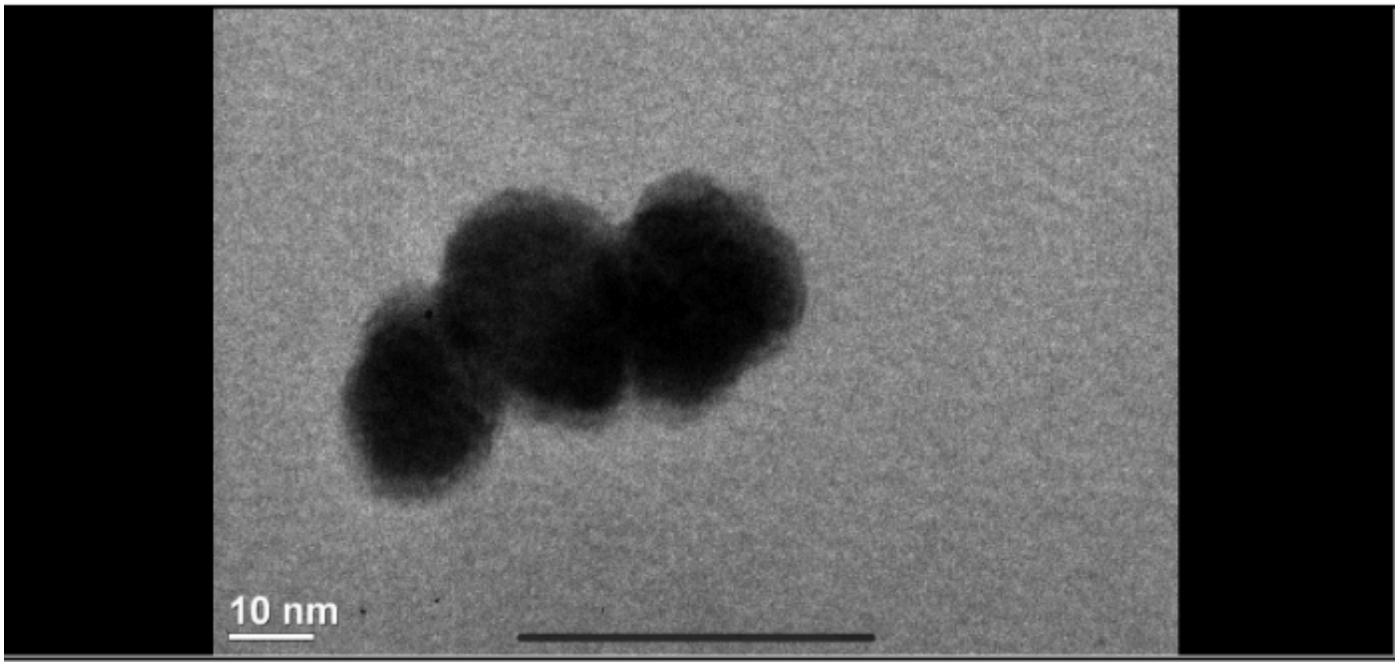
4. Design Sketches





The array housing is designed with laser cutting and minimal 3D printing in mind, to offer the fastest and cleanest final product. Using acrylic will be necessary for creating a sealed environment for the suspended MNPs. Everything will be bonded with acrylic glue or screwed together using threaded inserts.

The only other intricate fabrication considerations are the MNPs themselves. Using high-temperature organic-phase thermal decomposition, which allows for high control over nucleation, size, and shape, the nanoparticles we use can be tailored to our use case. This works by slowly reacting iron oxides and other ferrites in boiling solvents with surfactants such as oleic acid. When coated with oleic acid, the hydrophobic tails prevent recombination of the introduced magnetic material, resulting in individual superparamagnetic particles. Using traditional synthesis techniques to regulate flow rate and control temperature, the size and shape of these particles can be highly tuned within nanometers. The ones used in our prototype will be spherical, with diameters on the order of 100nm, then removed from the solvent bath and suspended in the fluid chamber within our device.



5. Software Requirements Specification (SRS)

5.1 Definitions, Abbreviations

None

5.2 Functionality

ID	Description
SRS-01	The system shall communicate with the two solenoid boards through the I2C bus to independently control all 15 electromagnets in the array
SRS-02	The system shall monitor Hall effect sensor readings to verify that the magnetic fields are present in the correct pattern when electromagnets are activated

ID	Description
SRS-03	The system shall read the current time from the RTC module through the I2C interface periodically
SRS-04	The system shall implement a time display mode in which the microcontroller converts the current time into a pattern of electromagnets turning on that moves the magnetic field into positions that digitally represent the time
SRS-05	The system shall implement an animation mode in which the ATmega328PB sequentially activates electromagnets to predefined patterns to generate an animation
SRS-06	The system shall detect button presses through hardware interrupts to switch between time display mode and animation mode

6. Hardware Requirements Specification (HRS)

6.1 Definitions, Abbreviations

MCU - Microcontroller Unit

6.2 Functionality

ID	Description
HRS-01	The device shall include a 3x5 array of electromagnets each capable of being activated independently
HRS-02	The electromagnets shall be controlled by two 8-channel solenoid driver boards that utilize MOSFETs and flyback diodes that allow high-current switching without drawing current directly from the GPIO pins of the MCU.
HRS-03	The system shall use a regulated 5V power supply capable of supplying at least 3.3A to power the electromagnets, MCU, sensors and display.
HRS-04	The system shall use an ATmega328PB MCU to coordinate control of the electromagnets, sensors, display and the RTC module
HRS-05	The system shall include a Hall effect sensor connected to the MCU through a GPIO pin with ADC channels capable of detecting magnetic field strength generated by the electromagnet array

ID	Description
HRS-06	The system shall include a real-time clock module connected to the MCU via I2C to keep accurate time keeping, checking the time once per second.
HRS-07	The system shall include two push buttons for selecting operating modes by interrupting the MCU and an LCD display connected through I2C to show the system status and current mode.

7. Bill of Materials (BOM)

For our materials, we first and foremost need 15 electromagnets, through which we will manipulate magnetic fields to move magnetic nanoparticles/ferrofluid. Along with these electromagnets, we will need the power management ICs to properly power and control them via the MCU. For user interaction, we will need buttons to control the operating modes of our electromagnet array and LCD with I2C-enabled control to display important information. As a test probe, we will also use a Hall effect sensor to ensure that our electromagnets are operating at expected capacities. Lastly, we will use a real-time clock chip to enable accurate tracking of the time.

https://docs.google.com/spreadsheets/d/1sWL6C1-NioSSsZ-AeGwEUh-nf6-uot8BXnSm6up-Bj0/edit?usp=sharing

8. Final Demo Goals

For the final demonstration, the device will be put on the table, with the electromagnetic array clearly visible. During the demo, we will feature the time display mode first by pressing the time display mode button waiting a couple of minutes to see the magnetic materials move when the minute changes to their new respective positions. Then the animation mode with the respective button as well, and the MCU will cycle through the specific shapes pre-programmed on the MCU.

9. Sprint Planning

Milestone	Functionality Achieved	Distribution of Work
Sprint #1	Electromagnet array assembled and validated by wiring up the 3x5 electromagnet array, connecting the solenoid driver boards, and ensuring that the magnetic material can be moved in a controllable and repeatable way without a catastrophic failure in the power supply, and that there is stability	Equal
Sprint #2	Integrate the Hall-effect sensor system and verify that the electromagnets' magnetic fields can be reliably measured by the Hall-effect sensors.	Equal
MVP Demo	Make the user interface system by wiring the LCD display, mode-selecting buttons, and the RTC chip, and then writing the software.	Equal
Final Demo	Fully fucntional magnetic field control prototype	Equal

This is the end of the Project Proposal section. The remaining sections will be filled out based on the milestone schedule.

Sprint Review #1

Last week's progress

Current state of project

Next week's plan

Sprint Review #2

Last week's progress

Current state of project

Next week's plan

MVP Demo

Final Report

Don't forget to make the GitHub pages public website! If you've never made a GitHub pages website before, you can follow this webpage (though, substitute your final project repository for the GitHub username one in the quickstart guide): https://docs.github.com/en/pages/quickstart

1. Video

2. Images

3. Results

3.1 Software Requirements Specification (SRS) Results

ID	Description	Validation Outcome
SRS-01	The IMU 3-axis acceleration will be measured with 16-bit depth every 100 milliseconds +/-10 milliseconds.	Confirmed, logged output from the MCU is saved to "validation" folder in GitHub repository.

3.2 Hardware Requirements Specification (HRS) Results

ID	Description	Validation Outcome
HRS-01	A distance sensor shall be used for obstacle detection. The sensor shall detect obstacles at a maximum distance of at least 10 cm.	Confirmed, sensed obstacles up to 15cm. Video in "validation" folder, shows tape measure and logged output to terminal.

ID	Description	Validation Outcome

4. Conclusion

References